

PS13 RESEARCH DOCUMENTATION

Hey guys this is basis documentation of our problem I research on all concepts that technology use in our project. I found that in our project given technology concepts uses like:-

- MPLS
- SD-WAN
- NOC
- Predictive Maintenance
- Air-Gapped Systems

Here is detailed breakdown of each concept I want you should check all concepts ones so we know what exactly we required in our project

1. MPLS (Multi-Protocol Label Switching)

What is MPLS?

MPLS is a private and secure networking technology used by large organizations to connect multiple offices and data centers.

Simple Example

Imagine ISRO has offices in:

- Bengaluru
- Ahmedabad
- Sriharikota

These offices need fast and secure communication.

Instead of using the public internet, they use MPLS.

Why MPLS?

- Faster communication
- Better security
- Stable network performance

- **Reliable routing**

In Our Project

We are NOT building MPLS.

We are building an AI Copilot that monitors MPLS networks and predicts failures before they happen.

2. SD-WAN (Software Defined Wide Area Network)

What is SD-WAN?

SD-WAN is a smart networking technology that automatically chooses the best path for network traffic.

Simple Example

Like Google Maps:

If one road has traffic, Google Maps chooses another route.

Similarly, SD-WAN chooses the best network path.

Benefits

- **Better performance**
- **Lower latency**
- **Automatic route optimization**
- **Reduced downtime**

In Our Project

Our AI can analyze network conditions and suggest better routing decisions when congestion is detected.

3. NOC (Network Operations Center)

What is NOC?

NOC is a centralized monitoring room where engineers continuously monitor network health.

What NOC Engineers Monitor

- **CPU Usage**
- **Memory Usage**
- **Latency**
- **Packet Loss**
- **Router Health**
- **Link Status**

Current Workflow

Engineer manually checks logs and dashboards.

Our Project Goal

Create an AI Copilot that assists NOC engineers by:

- **Monitoring logs**
- **Explaining problems**
- **Predicting failures**
- **Suggesting solutions**

4. Predictive Maintenance

What is Predictive Maintenance?

Predictive Maintenance means identifying problems before they actually happen.

Traditional System

Problem occurs



Alert generated

Predictive System

Warning signs detected



Future failure predicted



Action suggested

Example

CPU Usage Trend:

40%



60%



75%



90%

AI Prediction:

"Network congestion likely within the next 2 hours."

Why Important?

- **Reduces downtime**
- **Improves reliability**
- **Prevents major failures**

In Our Project

This is the most important feature.

Our AI will predict:

- **Network congestion**

- **Link failures**
- **Performance degradation**

before users are affected.

5. Air-Gapped Systems

What is an Air-Gapped System?

An Air-Gapped system is completely disconnected from the internet.

Why Use Air-Gapped Systems?

Organizations like:

- **ISRO**
- **Defence Agencies**
- **Military Networks**
- **Nuclear Facilities**

need maximum security.

Limitation

They cannot use:

- **ChatGPT**
- **Cloud AI**
- **Online Services**

Our Solution

Use Local AI Models such as:

- **Llama 3**
- **Mistral**
- **Gemma**

running through Ollama.

This allows AI assistance without internet access.

Project Vision

Current Situation:

Network Issue

↓

Engineer investigates manually

↓

Problem identified

Our Solution:

Network Logs

↓

AI Copilot

↓

Failure Prediction

↓

Reason Explanation

↓

Recommended Action

Final Project Features

- 1. Network Log Upload**
- 2. AI Risk Prediction**
- 3. Failure Probability Score**
- 4. Root Cause Explanation**
- 5. Suggested Troubleshooting Actions**
- 6. Offline AI Support**

7. NOC Dashboard

One-Line Summary

AlphaCopilot is an offline AI-powered Network Operations Assistant that helps engineers predict network failures, understand root causes, and take preventive action before operational impact occurs.